GameFlow Traffic Control (GTC)

A Game Theory Approach To Traffic Signal Optimization

Group Members

Syed Subhan Shah Bukhari (22P-9447)

Maaz Mustafa (22P-9405)

Moiz Ghazanfar(22P-9425)

Project Supervisor

Mr. Usama Musharaf

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Problem Statement

“Conventional traffic signals use fixed-time cycles, which are inefficient as they cannot adapt to real-time traffic fluctuations, leading to unnecessary congestion and long delays.”

Literature Review: Performance Evaluation

Method	Deployment Status	Evidence
🏆 Game Theory	✅ PROVEN - 350+ Cities	• SCOOT: 350+ cities worldwide (1976-present) • SCATS: Widespread US deployment • Los Angeles: 457 intersections • Blacksburg, VA: 38 intersections
Reinforcement Learning	❌ ZERO Deployments	Chen et al. (2022): *"RL-based signal controllers have never been deployed"*
Genetic Algorithms	❌ ZERO Deployments	All research confined to simulators *"No way to perform feedback in real world"*

Literature Review: Performance Comparison

Criterion	Game Theory	RL	GA
Convergence Speed	⚡ 10-50 iterations (seconds)	⌛ Months of training	⌛ 100s-1000s generations
Real-Time Capable	✅ YES	❌ NO	❌ NO
Stability Guarantee	✅ Nash Equilibrium	❌ None	❌ None
Safety	✅ Guaranteed	⚠️ Trial-and-error unsafe	⚠️ No guarantees
Interpretability	✅ Clear payoffs	❌ Black-box	❌ Black-box

The ‘Black Box’ Problem

Reinforcement Learning:

“Complex and opaque neural networks hard to understand and debug" (Chen et al., 2022)

Cannot explain why a decision was made

Safety-critical systems require transparency

Genetic Algorithms:

"Survival of the fittest" is not an engineering explanation

No clear decision logic

Difficult to validate and tune

Game Theory:

Clear payoff functions: $U = -(\alpha \cdot \text{Wait} + \beta \cdot \text{Delay} + \gamma \cdot \text{Queue})$

Engineers understand WHY decisions are made

Easy to validate and adjust

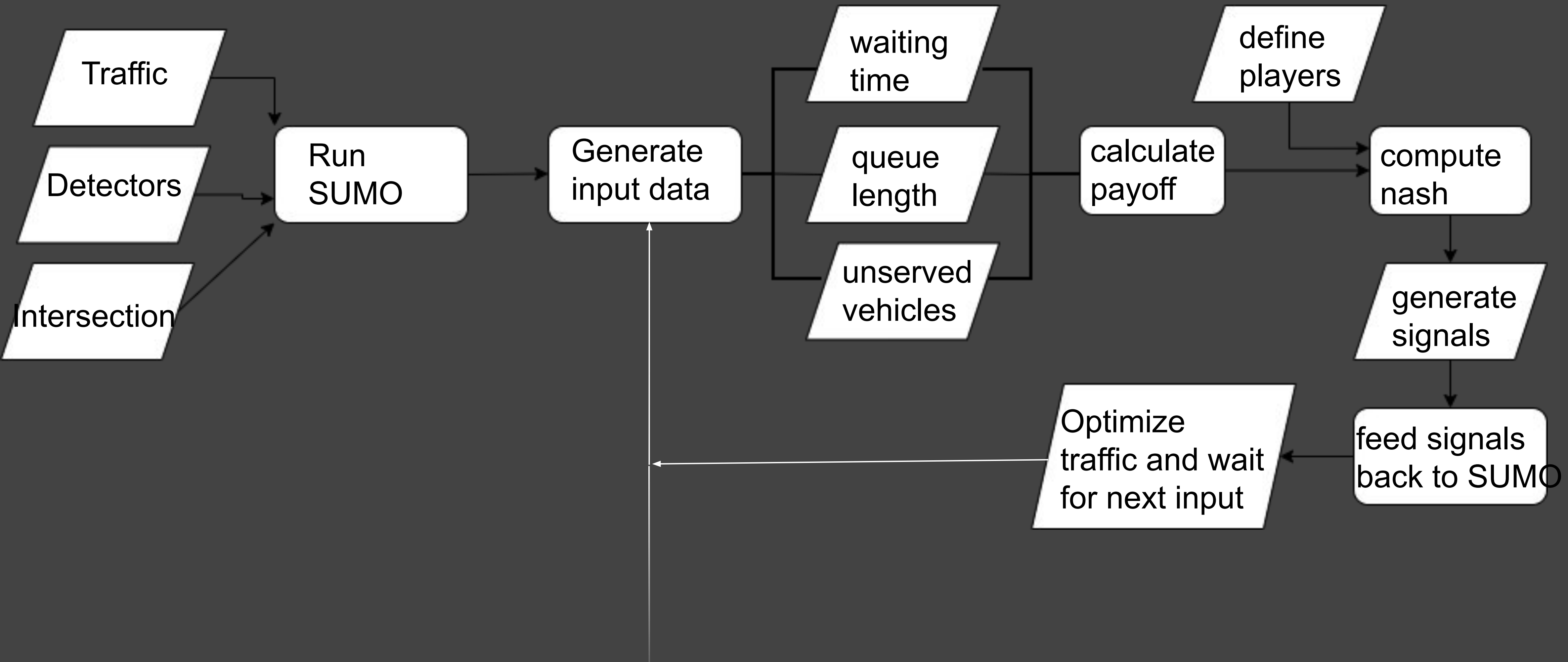
Timeline & Our Contribution

- 1970s ———— SCOOT/SCATS Development (Game Theory-based)
- 1976 ———— First SCOOT Deployment (Glasgow)
- 1980s ———— SCATS Deployment (Australia → US)
- 2000s ———— GA Research Begins (simulation only)
- 2010s ———— RL Research Surge (simulation only)
- 2019 ———— Zaied et al.: 457 intersections (Game Theory)
- 2022 ———— Chen et al. Review: "RL never deployed"
- 2023 ———— Wang et al.: Queue Equilibrium (Game Theory)
- 2025 ———— Our FYP: Game Theory in Pakistan

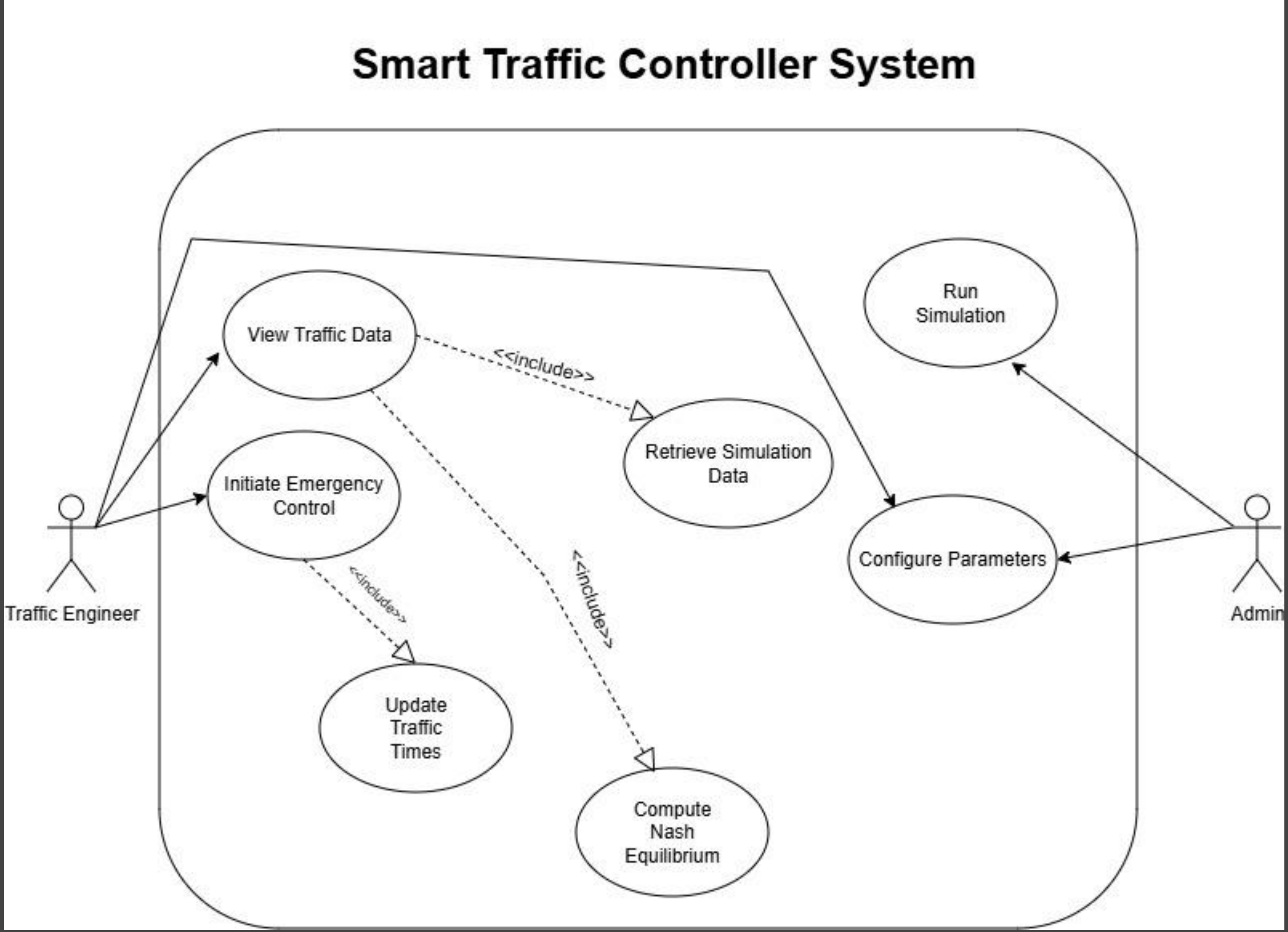
Our Adaptation:

- Modified for multiple timeslots
- Calibrated for mixed traffic
- 2-phase system (NS + EW)
- Nash Equilibrium optimization
- SUMO simulation validation

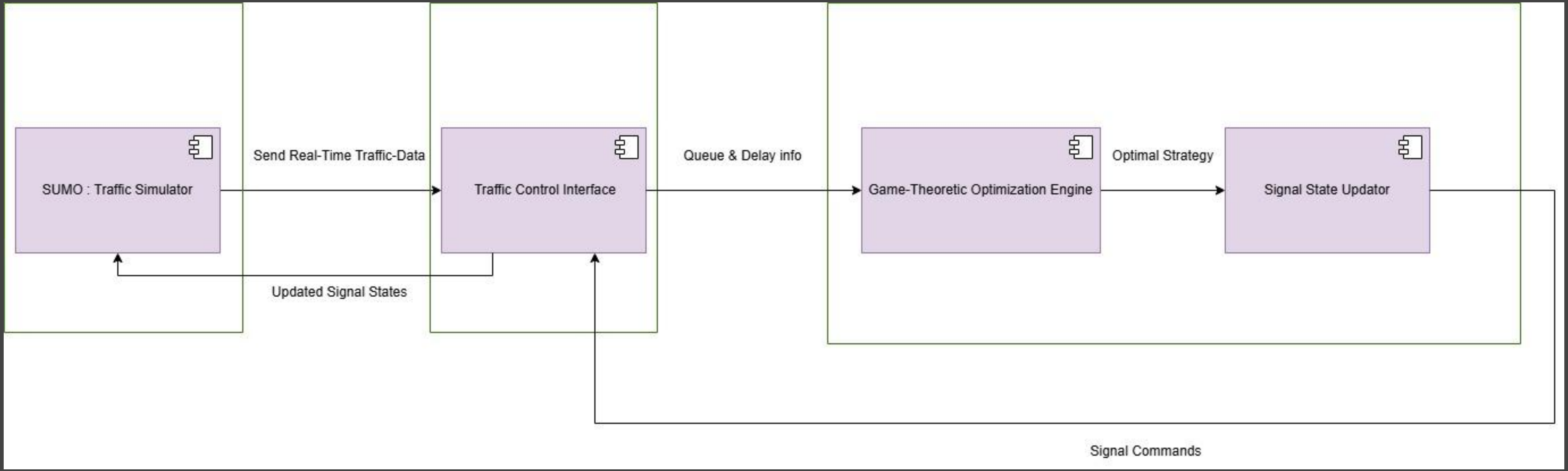
Updated Workflow Diagram



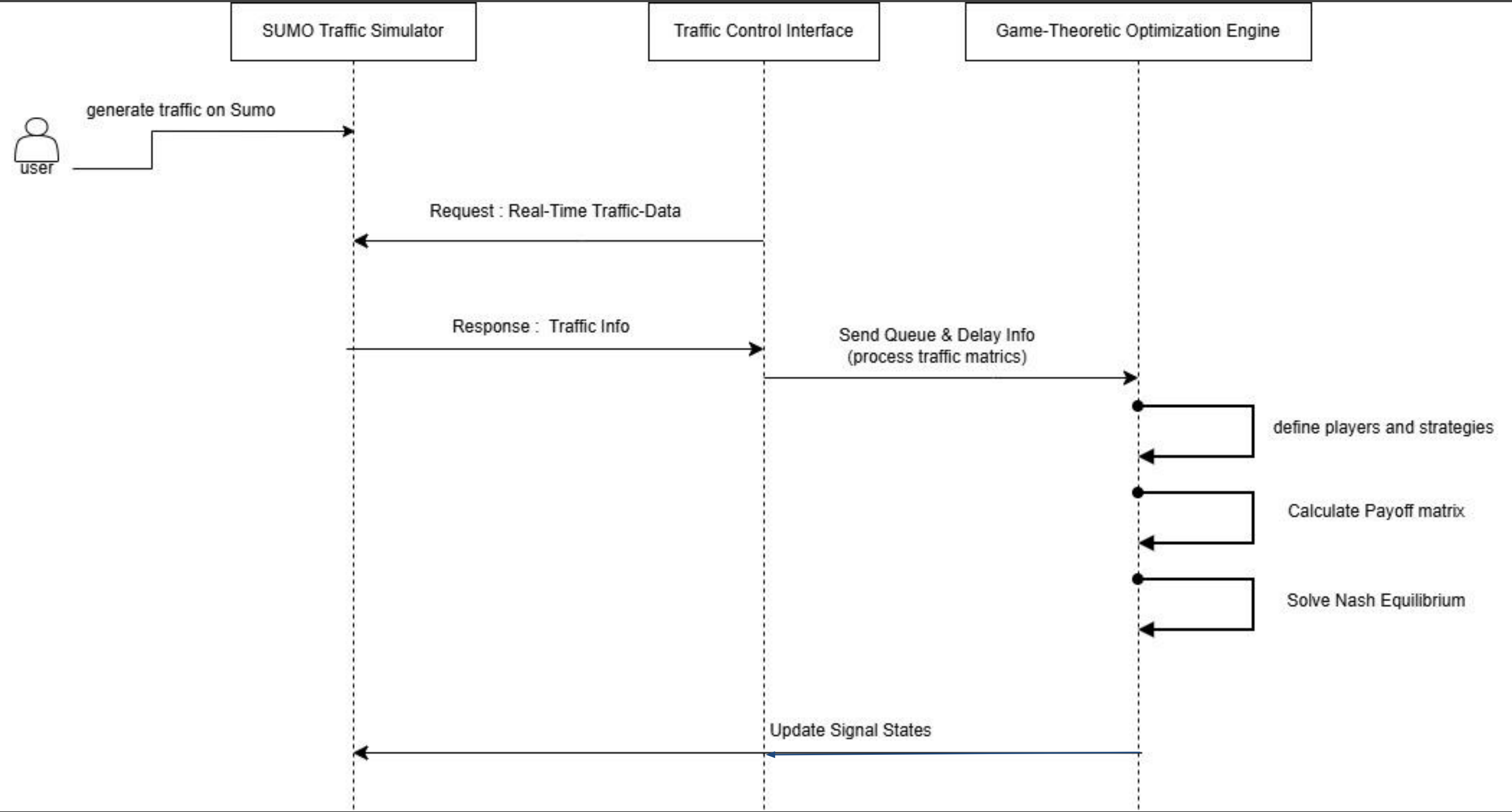
Updated Use-Case Diagram



Architecture Diagram



Activity Diagram



SUMO and TraCI

- Dynamic Network Generation
- High-Fidelity Mixed-Traffic Physics
- TraCI-Based Stochastic "Chaos”
- Dynamic Temporal Phases & Analytics

Fixed Timed Signals

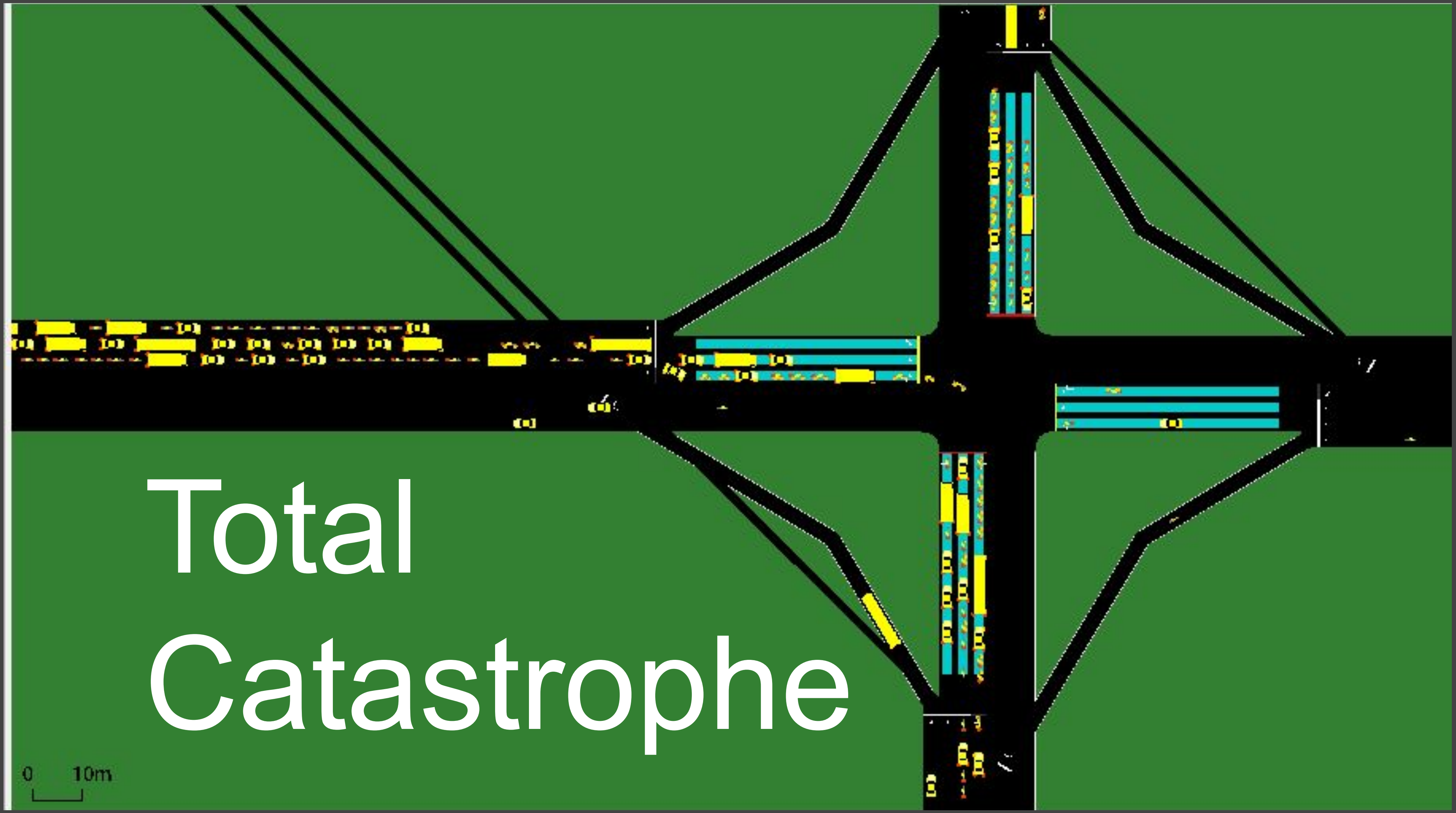
(Base Model)

Version 1

Constant Variables:
Type of Traffic: Mix and Aggressive
Time Slots = {Early Morning, Afternoon, Evening}
Number of Vehicles per each direction = 10200
Strategy Set = {Red, Yellow, Green}
Number of players = 2

Changing Variables for Version 01:
Sliplanes
Payoff = (α)QueueLength

Fixed Timed Model



Fixed Timed Model

North & South

Speed: 0.03

Travel Time: 1132.37 seconds.

Teleporting: 36

West & East

Speed: 0.40 to 0.47

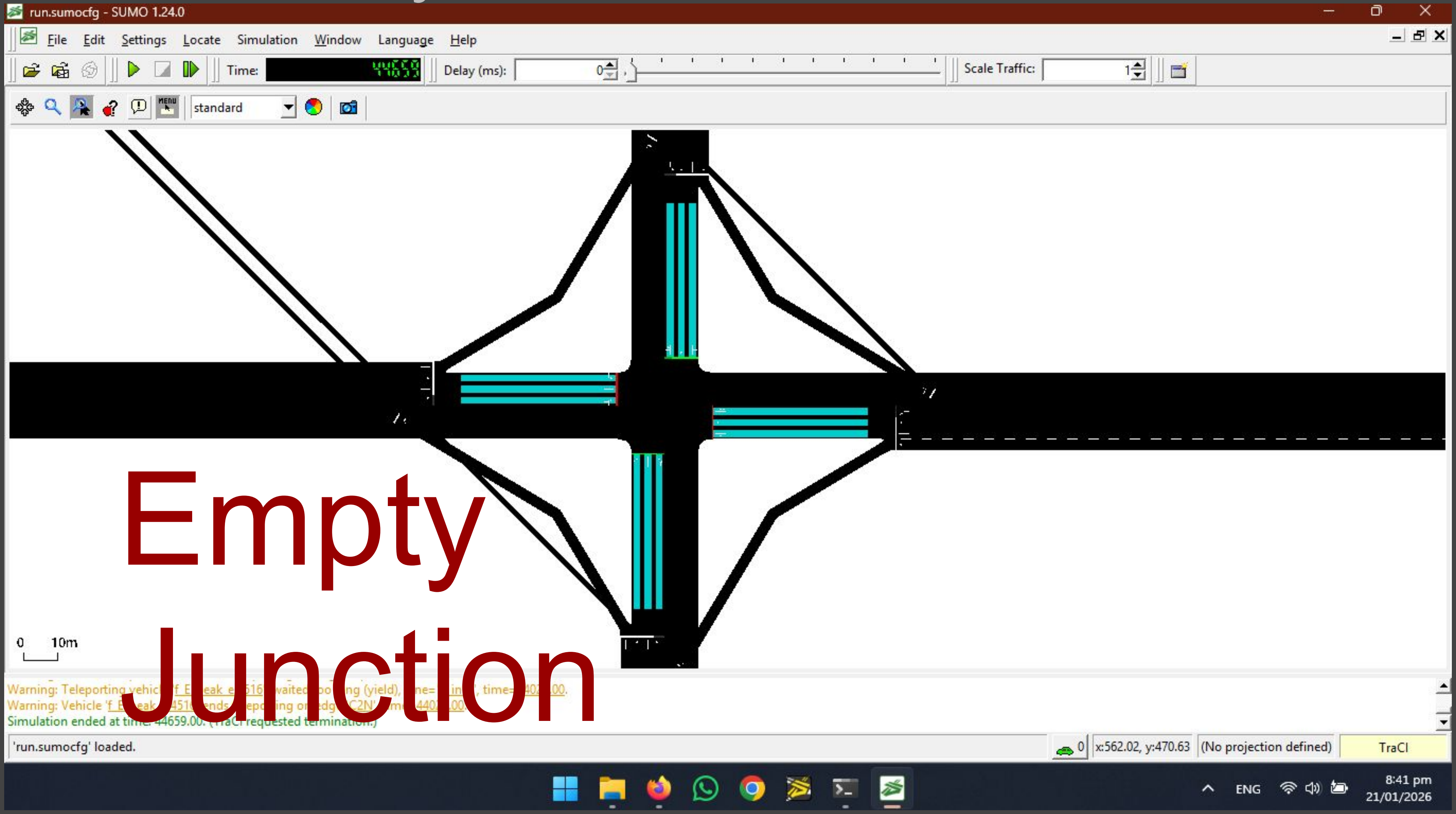
Travel Time: ~65–75 seconds.

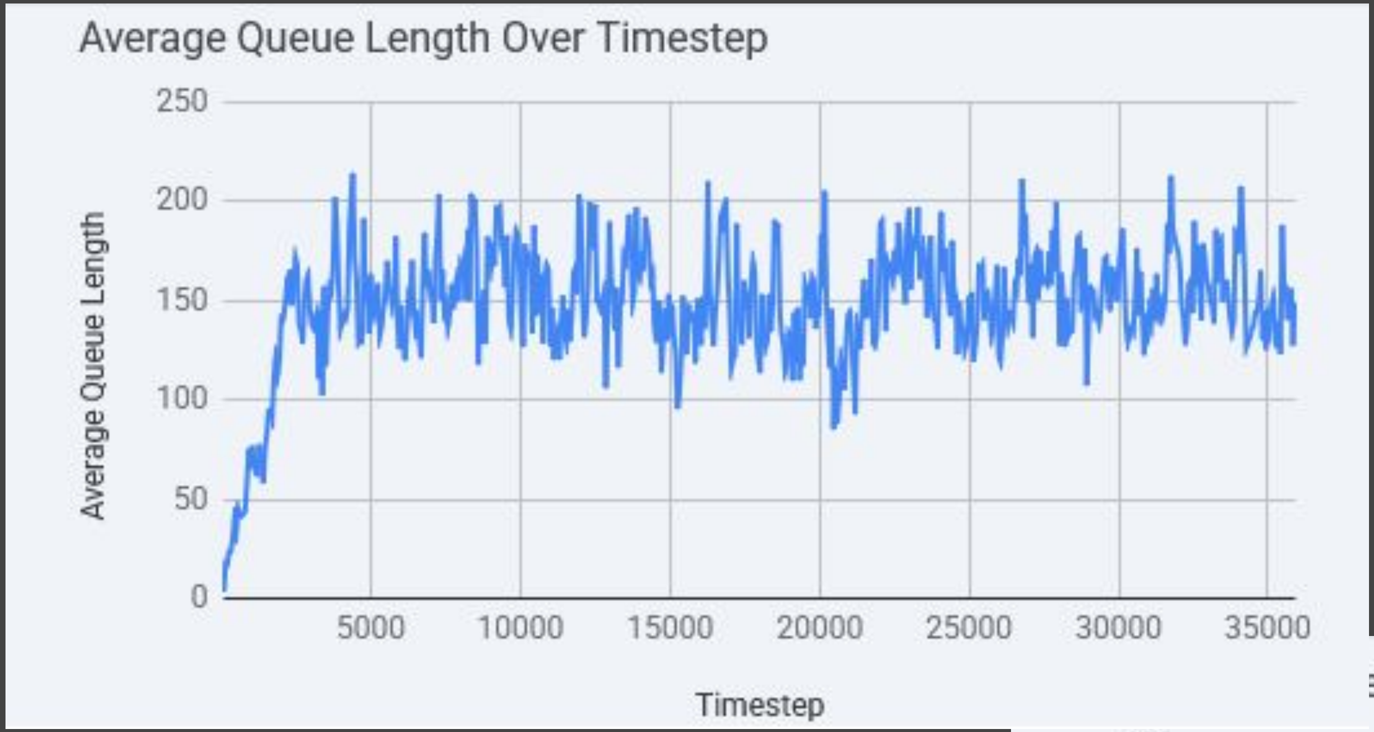
Spillback along with Unfairness!

Game Theory Solution

(With Slip Streams)

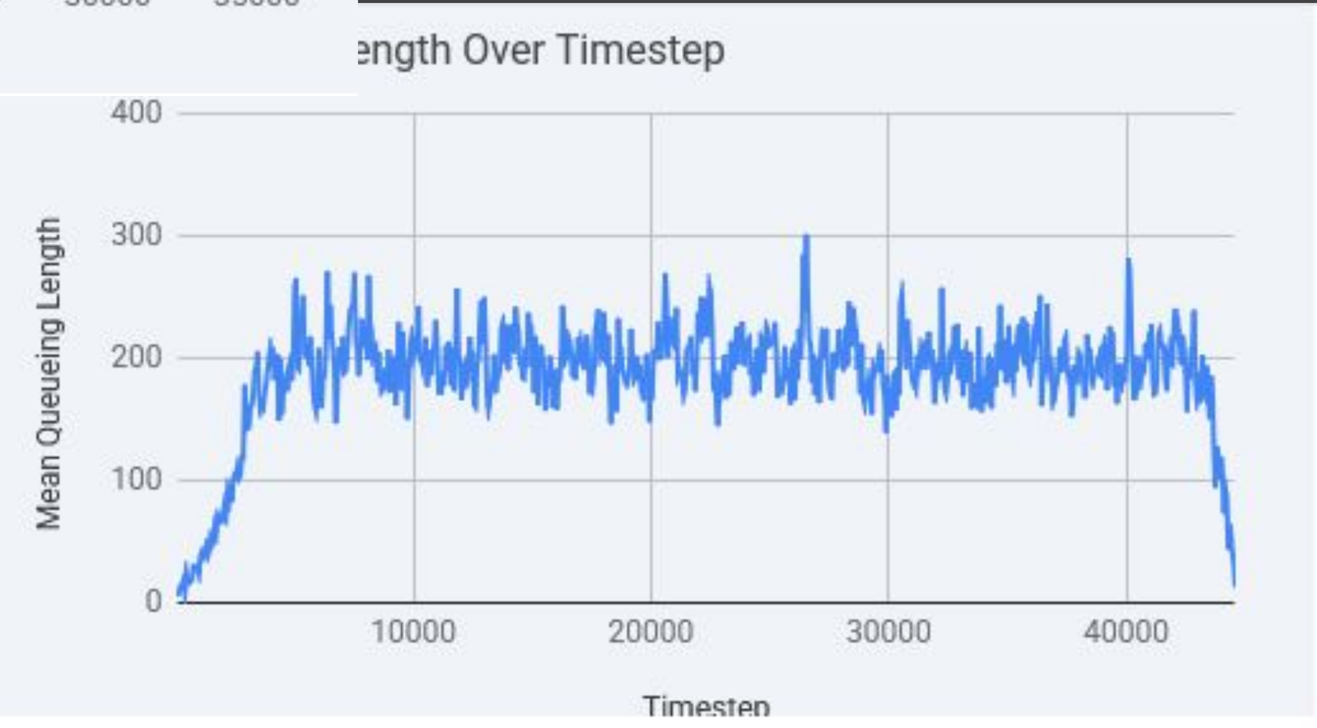
Game Theory Model





Fixed Timed

Game Theory Model



Analysis of GT Solution and Comparison

Average Waiting Time = ~ 500 sec

Average Queue Time = 37 sec

Game Theory was the Solution but had more average waiting time(base model has ~360), WHY?

“Survivor Bias”

Survivor Bias

Road (Edge)	Baseline (Fixed) Cars Passed	Game Theory Cars Passed	Improvement
North (`N2C`)	6,022	**10,200**	**+69% (Huge!)** 🚀
West (`W2C`)	6,353	**10,200**	**+60%**
South (`S2C`)	10,182	**10,200**	+0.1% (Maintained)
East (`E2C`)	10,188	**10,200**	+0.1% (Maintained)

But its NOT ENOUGH!

Version 2

Constant Variables:

Type of Traffic: Mix and Aggressive

Time Slots = {Early Morning, Afternoon, Evening}

Number of Vehicles per each direction = 10200

Strategy Set = {Red, Yellow, Green}

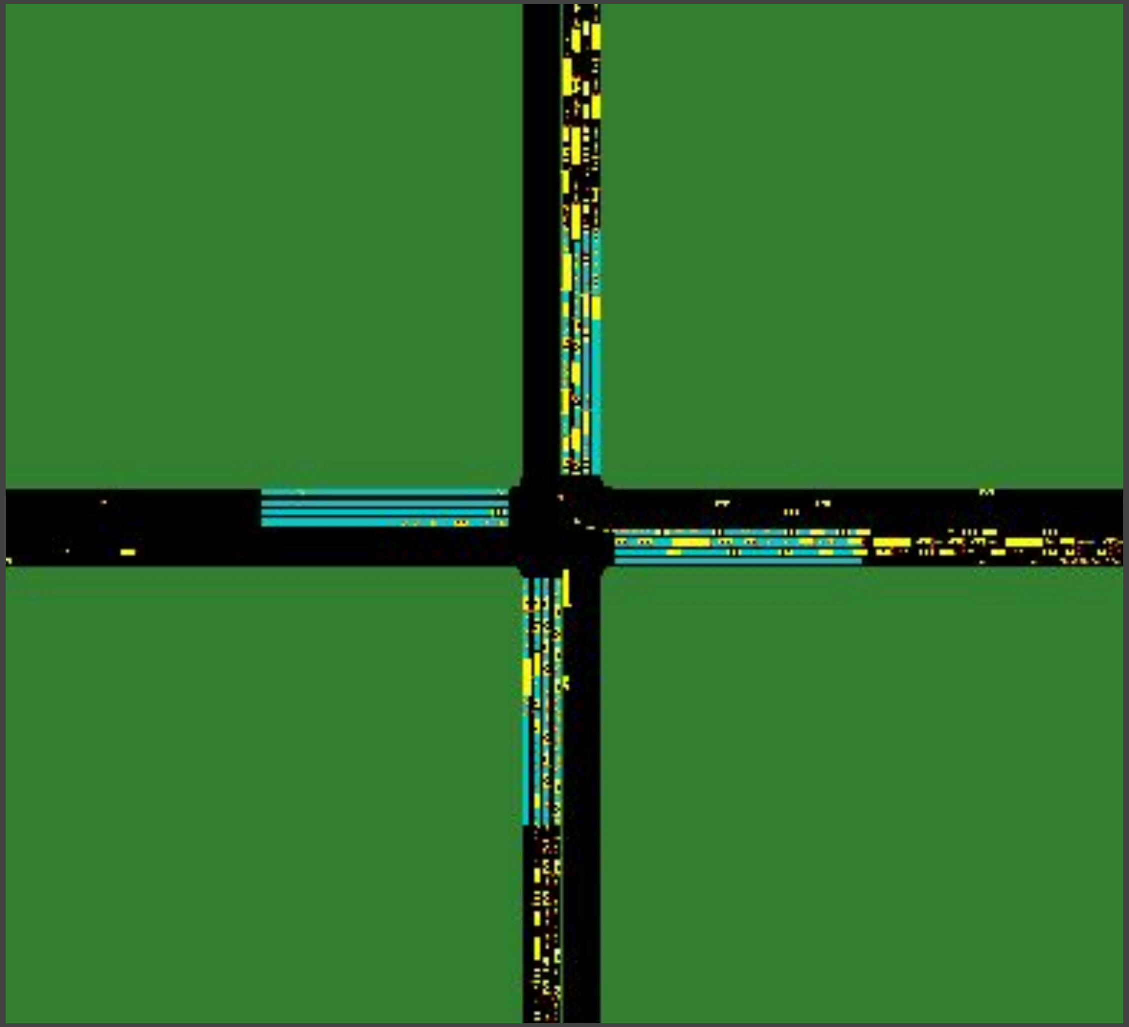
Number of player = 2

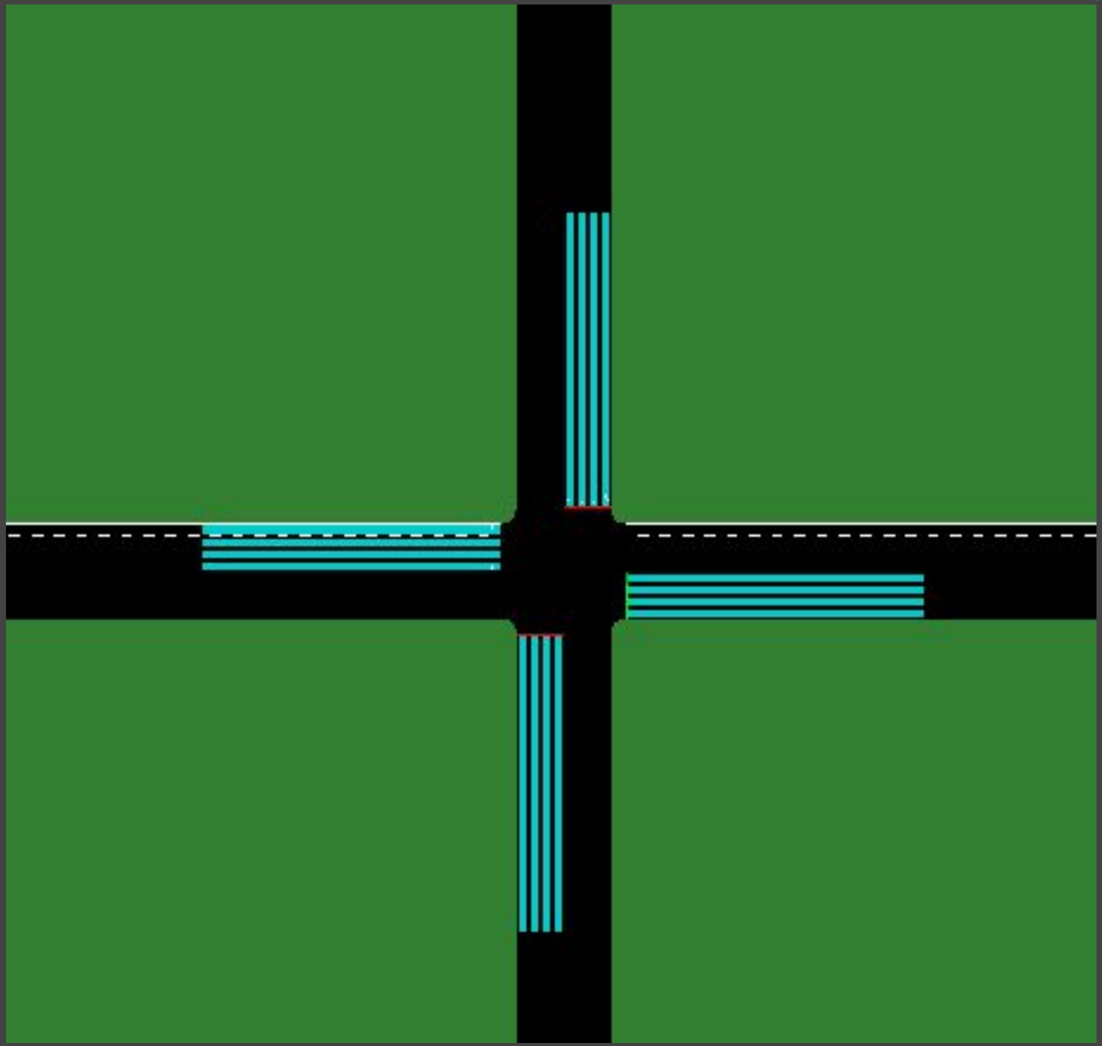
Changing Variables for Version 01:

No Sliplanes

Payoff = (α)QueueLength + (β)WaitingTime + (γ)ResidualVehicles

GT vs Base model Using Real Payoff





Gametheory at Work!

```
T=37107: Switching to NS. (NS Score: 928.0 vs EW Score: 447.0)
T=37129: Switching to EW. (NS Score: 886.5 vs EW Score: 980.5)
T=37144: Switching to NS. (NS Score: 1416.5 vs EW Score: 446.5)
T=37169: Switching to EW. (NS Score: 579.0 vs EW Score: 658.5)
T=37184: Switching to NS. (NS Score: 1144.5 vs EW Score: 148.0)
T=37219: Switching to EW. (NS Score: 398.0 vs EW Score: 440.0)
T=37234: Switching to NS. (NS Score: 476.5 vs EW Score: 8.0)
T=37278: Switching to EW. (NS Score: 103.5 vs EW Score: 120.5)
T=37293: Switching to NS. (NS Score: 212.0 vs EW Score: 0.0)
T=37363: Switching to NS. (NS Score: 672.0 vs EW Score: 0.0)
T=37433: Switching to NS. (NS Score: 408.5 vs EW Score: 0.0)
T=37503: Switching to NS. (NS Score: 293.0 vs EW Score: 0.0)
T=37573: Switching to NS. (NS Score: 105.0 vs EW Score: 0.0)
T=37643: Switching to NS. (NS Score: 58.0 vs EW Score: 0.0)
T=37713: Switching to NS. (NS Score: 79.5 vs EW Score: 0.0)
T=37783: Switching to NS. (NS Score: 31.0 vs EW Score: 0.0)
T=37853: Switching to NS. (NS Score: 42.0 vs EW Score: 0.0)
T=37923: Switching to NS. (NS Score: 60.0 vs EW Score: 0.0)
T=37993: Switching to NS. (NS Score: 44.0 vs EW Score: 0.0)
```

GT vs Base model Using Real Payoff

```
speed="0.96"
' speed="0.61"
speed="1.32"
speed="1.06"
```

```
speed="1.27"
speed="0.90"
speed="3.33"
speed="1.53"
```

GT vs Base model Using Real Payoff

Throughput

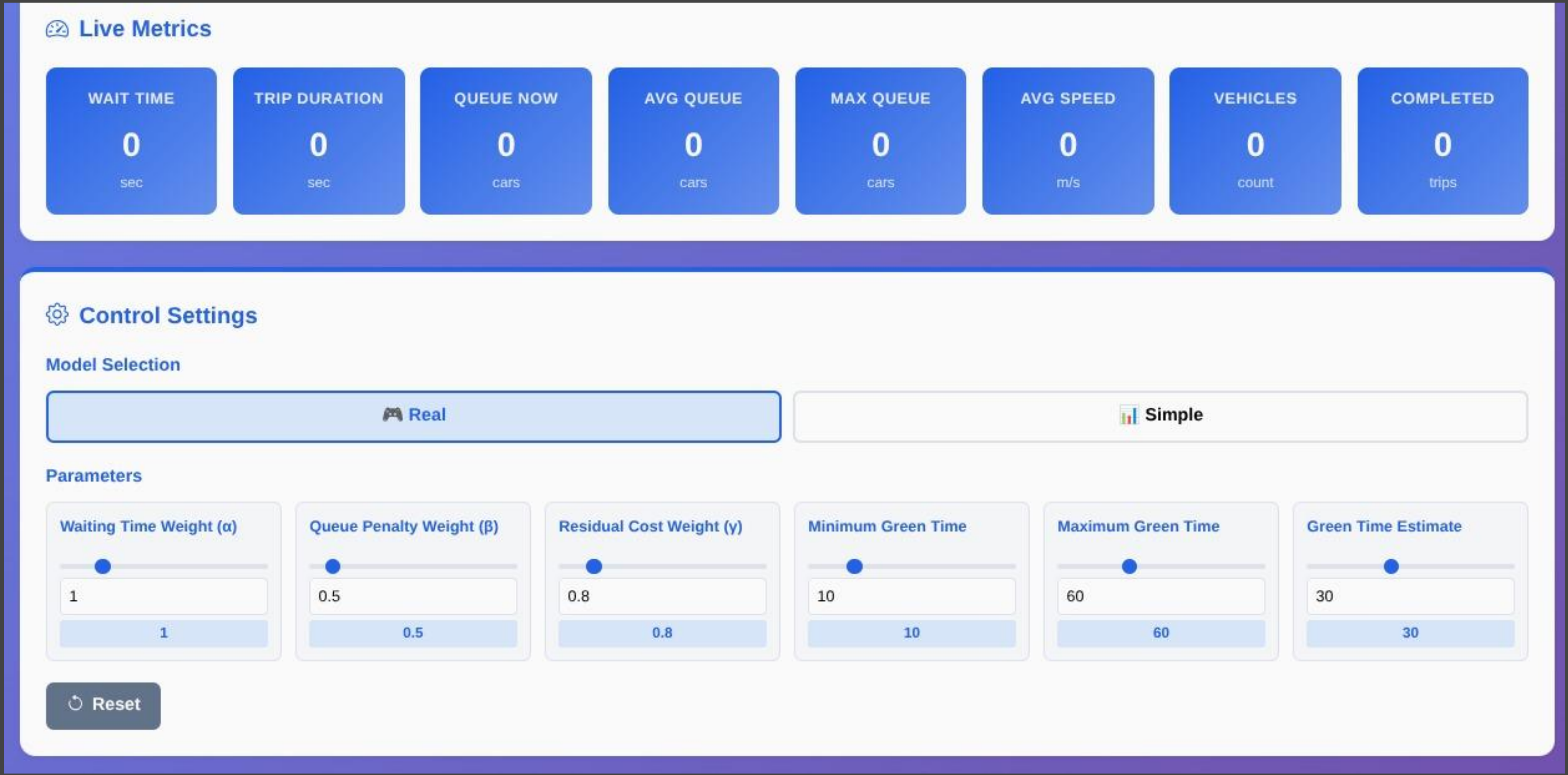
```
departed="8765"
' departed="9353"
departed="10048"
departed="10200"
```

```
departed="10200"
departed="10200"
departed="10200"
departed="10200"
```

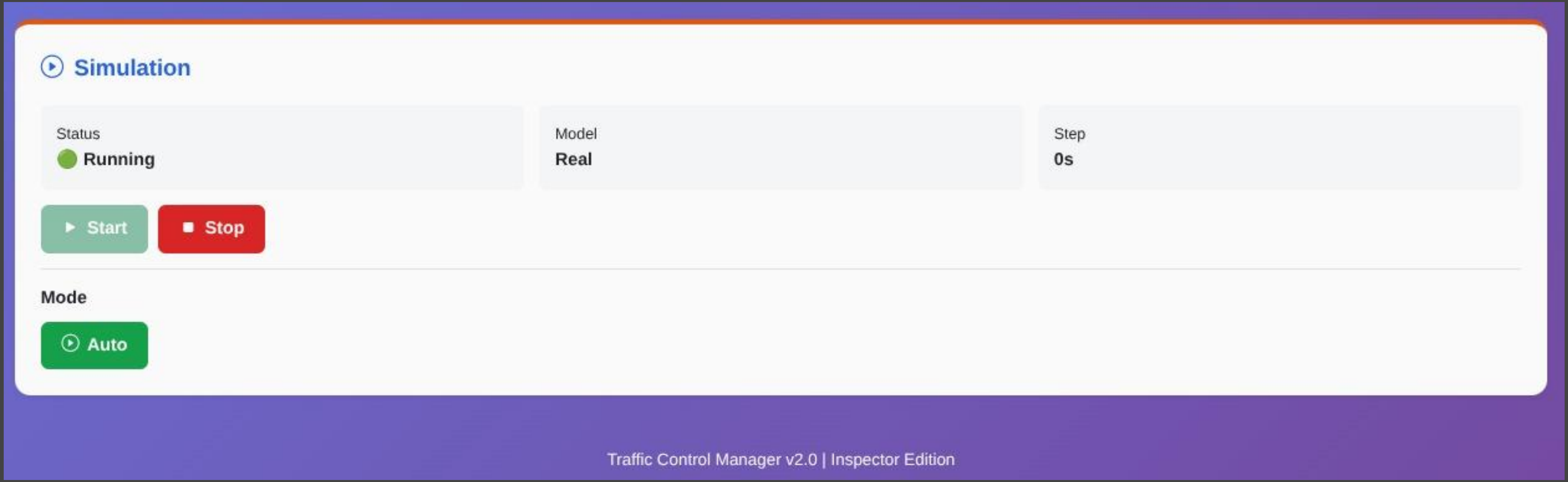
Avg Wait Time

Game Theory beat Fixed Timed Signals by half!!!

Project App Prototype



Project App Prototype



Project Planning

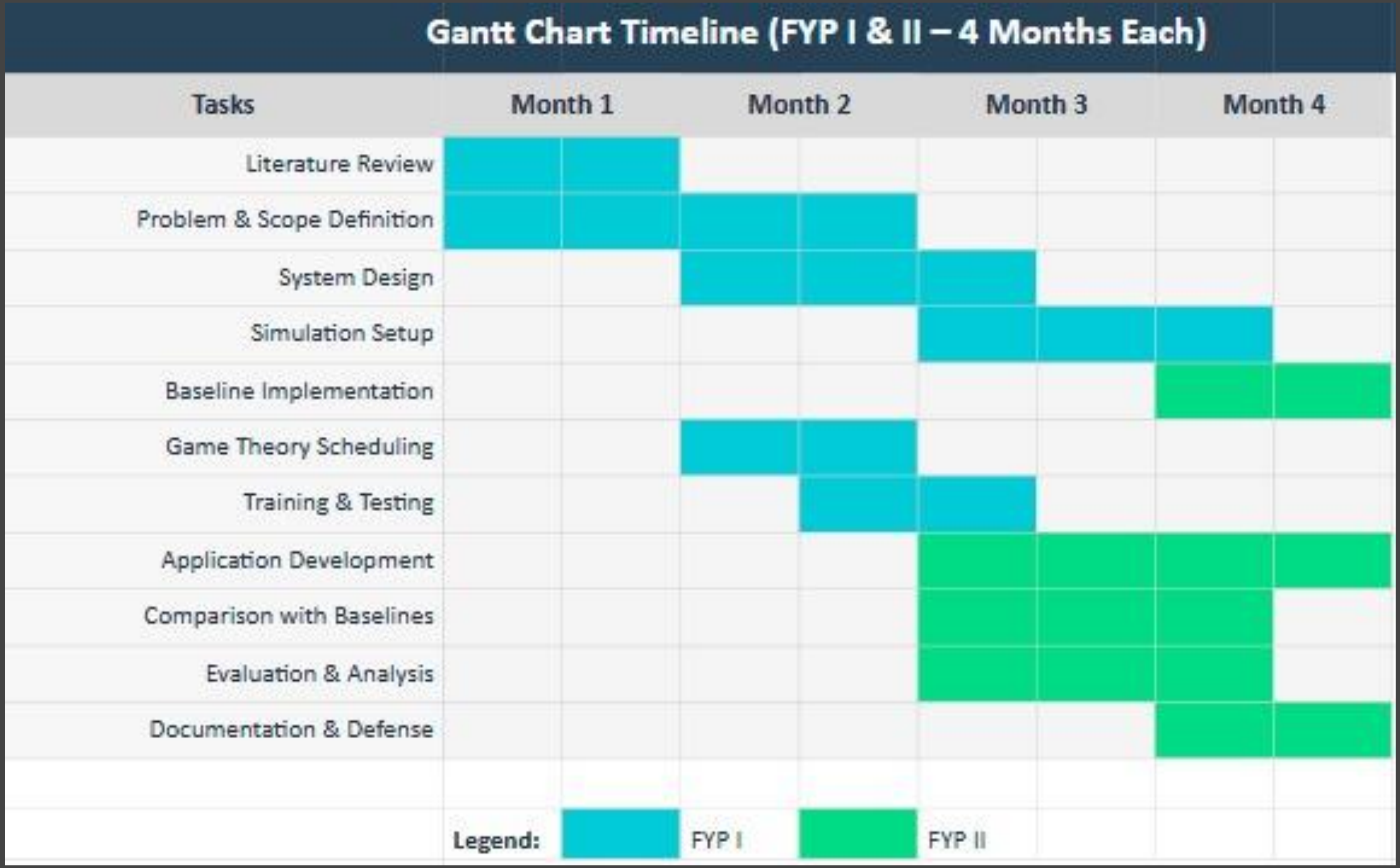


Fig: Gantt Chart

References

Game Theory:

1. **Zaied et al. (2019)** - *Sensors* - 457 intersections, Los Angeles
2. **Wang et al. (2023)** - *Applied Sciences* - Queue equilibrium control
3. **Stevanovic et al. (2009)** - SCOOT/SCATS deployment review
4. **TRL Software (2024)** - SCOOT: 350+ cities worldwide

Reinforcement Learning:

5. **Chen et al. (2022)** - *arXiv* - Comprehensive review
    - **Key Finding:** *"RL-based signal controllers have never been deployed"*
    - Identifies 4 insurmountable deployment barriers

Genetic Algorithms:

6. **Brazilian NSGA2 Study** - *"No way to perform feedback in real world"*
7. Multiple IEEE/ACM papers (2000-2020) - All simulation-based

Thank You!